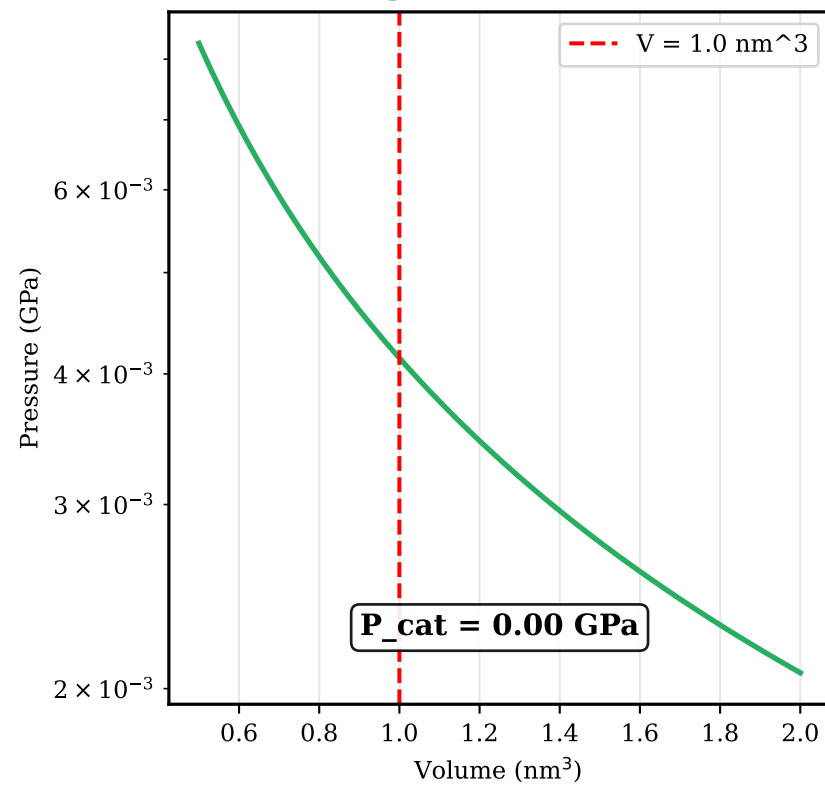
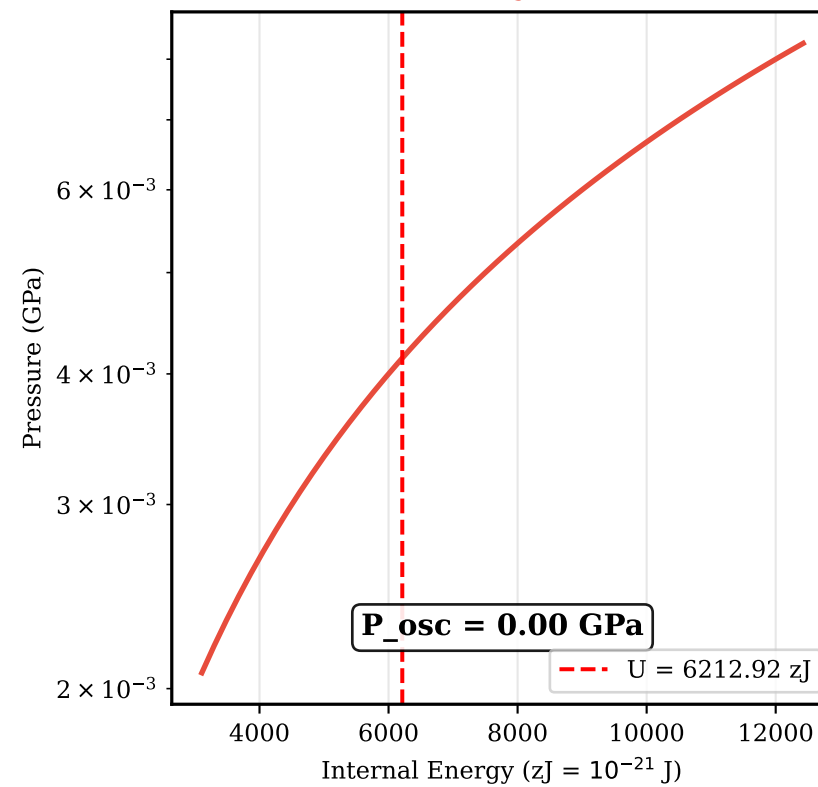


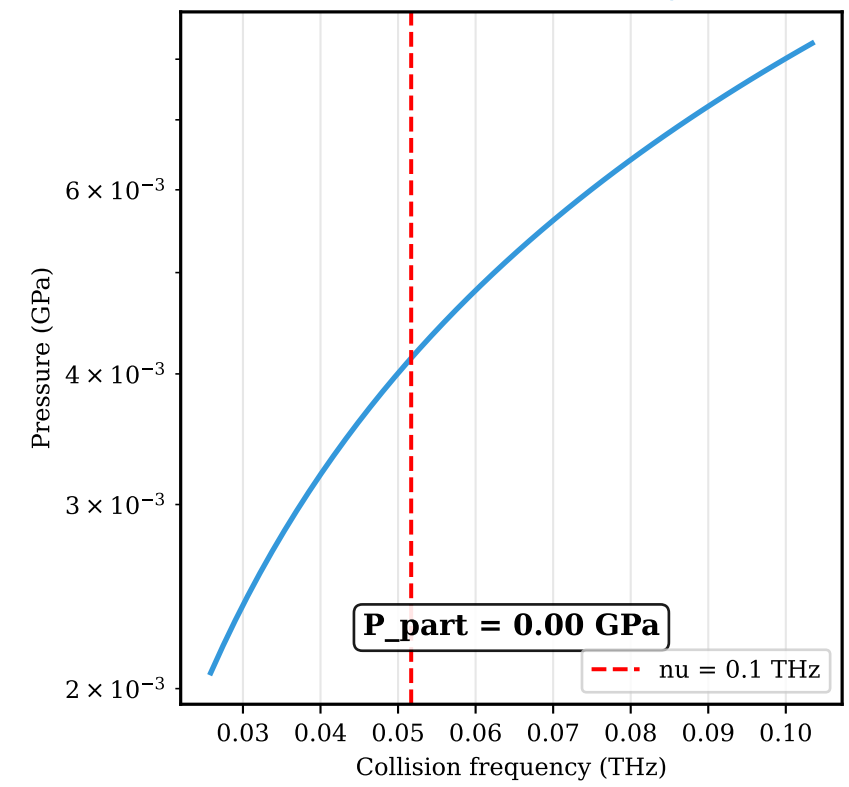
(A) Categorical: $P = k_B T (\partial M / \partial V)$



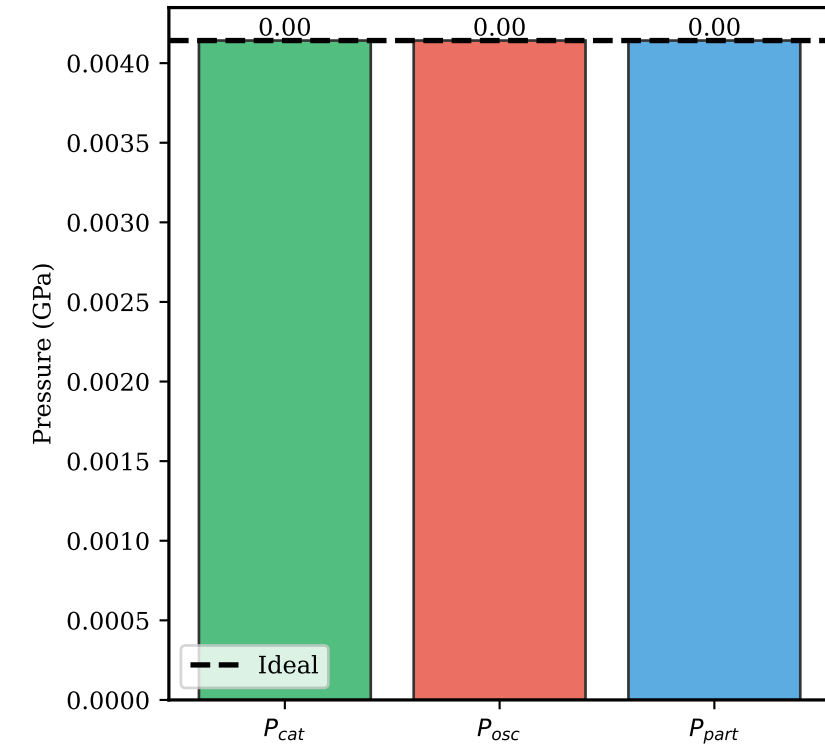
(B) Oscillatory: $P = 2U/(3V)$



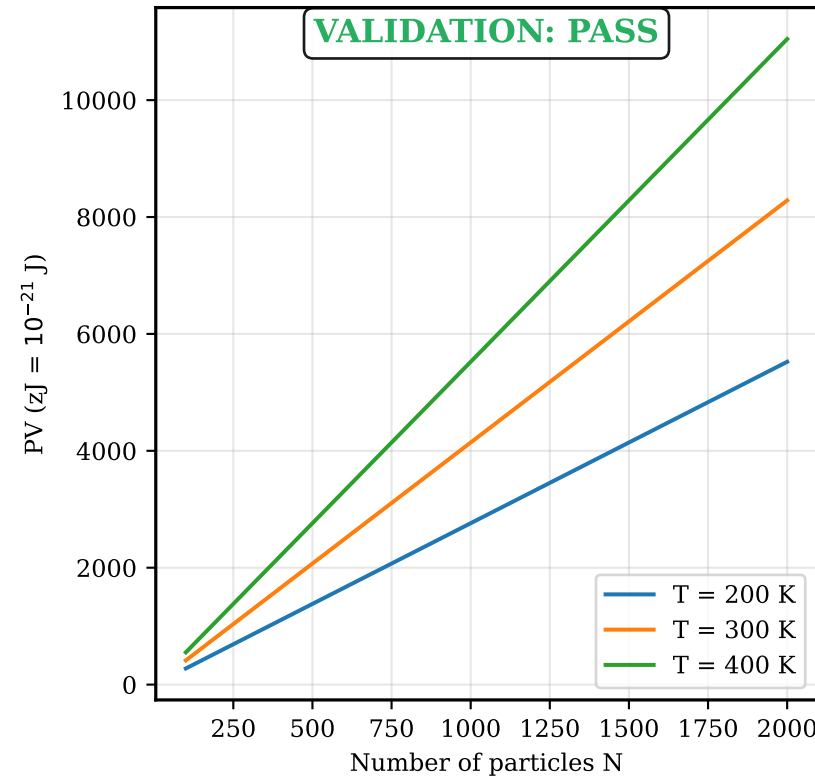
(C) Partition: $P = Nv\Delta p/A$



(D) Pressure Comparison



(E) $PV = NkT$ Verification



(F) Pressure Validation

PRESSURE EQUIVALENCE VALIDATION

Test: $P_{\text{cat}} = P_{\text{osc}} = P_{\text{part}} = NkT/V$

Parameters:

- $N = 1000$ particles
- $V = 1.0 \text{ nm}^3$
- $T = 300 \text{ K}$

Computed Pressures:

- $P_{\text{cat}} = 0.0041 \text{ GPa}$
- $P_{\text{osc}} = 0.0041 \text{ GPa}$
- $P_{\text{part}} = 0.0041 \text{ GPa}$

Ideal Gas Law:

- $PV = 4141.9470 \text{ zJ}$
- $NkT = 4141.9470 \text{ zJ}$
- Difference: 0.000000%

VALIDATION: PASS

CONCLUSION: Pressure is categorical density.
All three perspectives yield $PV = NkT$.